

DBMS ASSIGNMENT - 1

24MAB0A42

1. University Course Registration

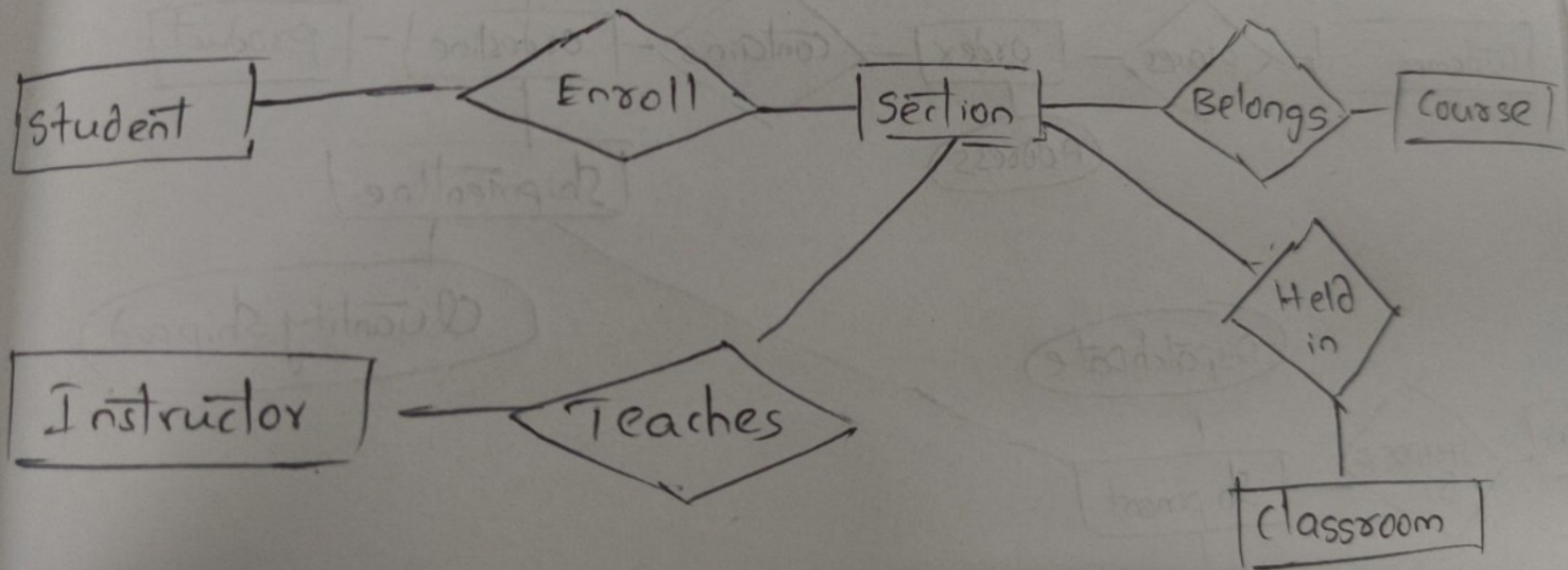
Flaws:

As given in the rules that a course can have different sections and each course has a separate instructor. Hence, there should be an entity named “section” and student enrolls in the section and each course has sections.

1. Semester and Year are attributes of Section as the same course is offered in many sections and years and if they are the attributes of Course, we might lost the data of the different semesters and years the course is offered.
2. Student should enroll in section and not in Course as the student registers for a section and not a course.
3. Since different sections of the Course can be taught by different instructors. Hence, Instructor teaches a section and not a Course.
4. Classroom allocation may differ among sections and hence a particular section is held in a classroom and not a Course.

University Course Registration

Correct ER-Diagram-



2. Hospital Prescription System

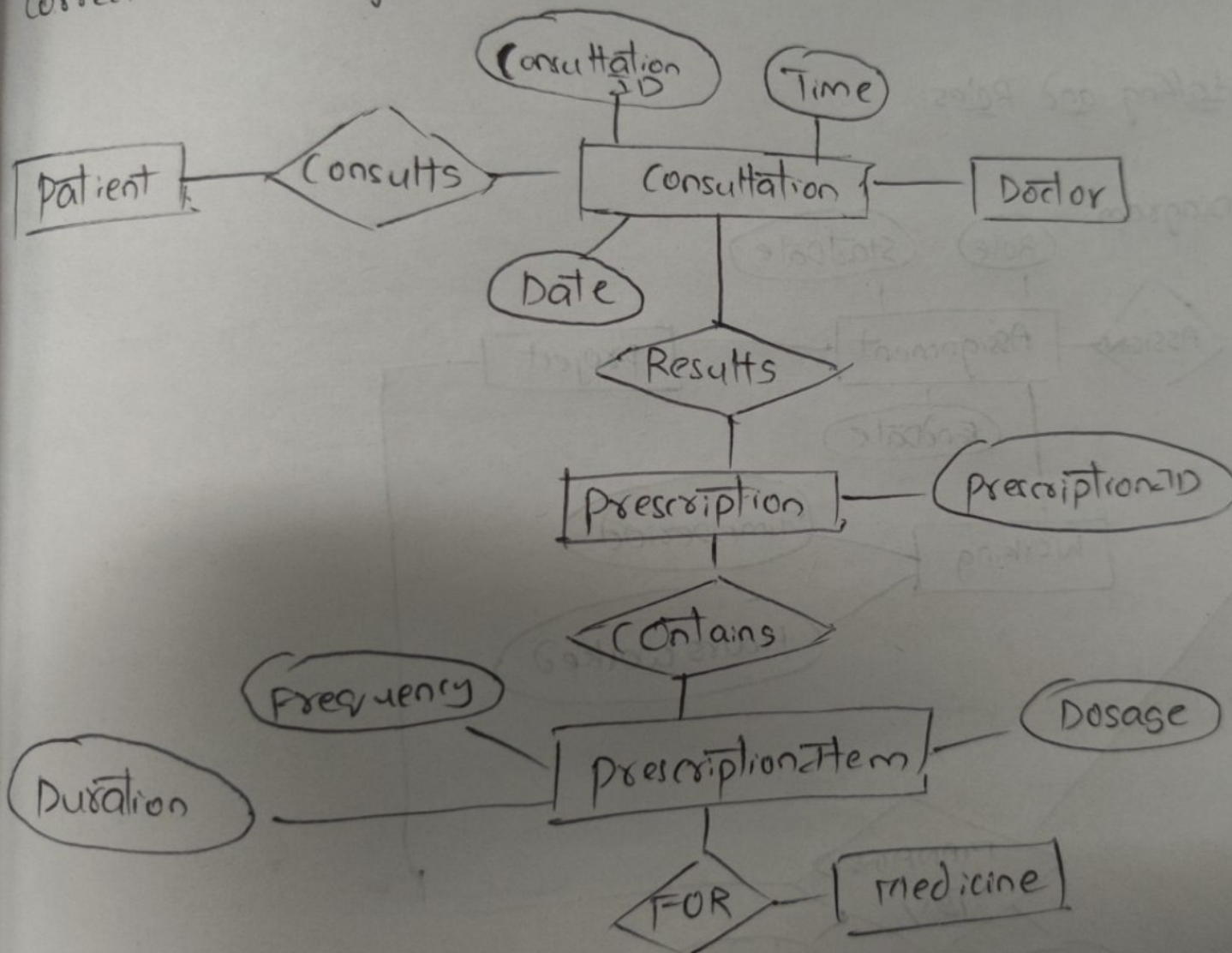
Flaws:

Since we need to store the consultation history and a patient might consult the same doctor many times, we need an entity "consultation", also one consultation results in "prescription" and that prescription will have certain "prescriptionItem" because the dosage is the attribute of prescriptionItem not the medicine.

1. Consultation should be an entity -> Patient ----CONSULTS----Consultation-----Doctor
2. Prescription should be an entity as well to keep the history whenever a patient consults a doctor.
3. PrescriptionItem is an entity which stores dosage as same medicine can be given in different dosages to two different individuals.
4. For time dependent instructions, we can put Frequency and Duration as attributes for PrescriptionItem.

a) Hospital Prescription System :-

Correct ER-Diagram -



3. E-Commerce Order Fulfilment

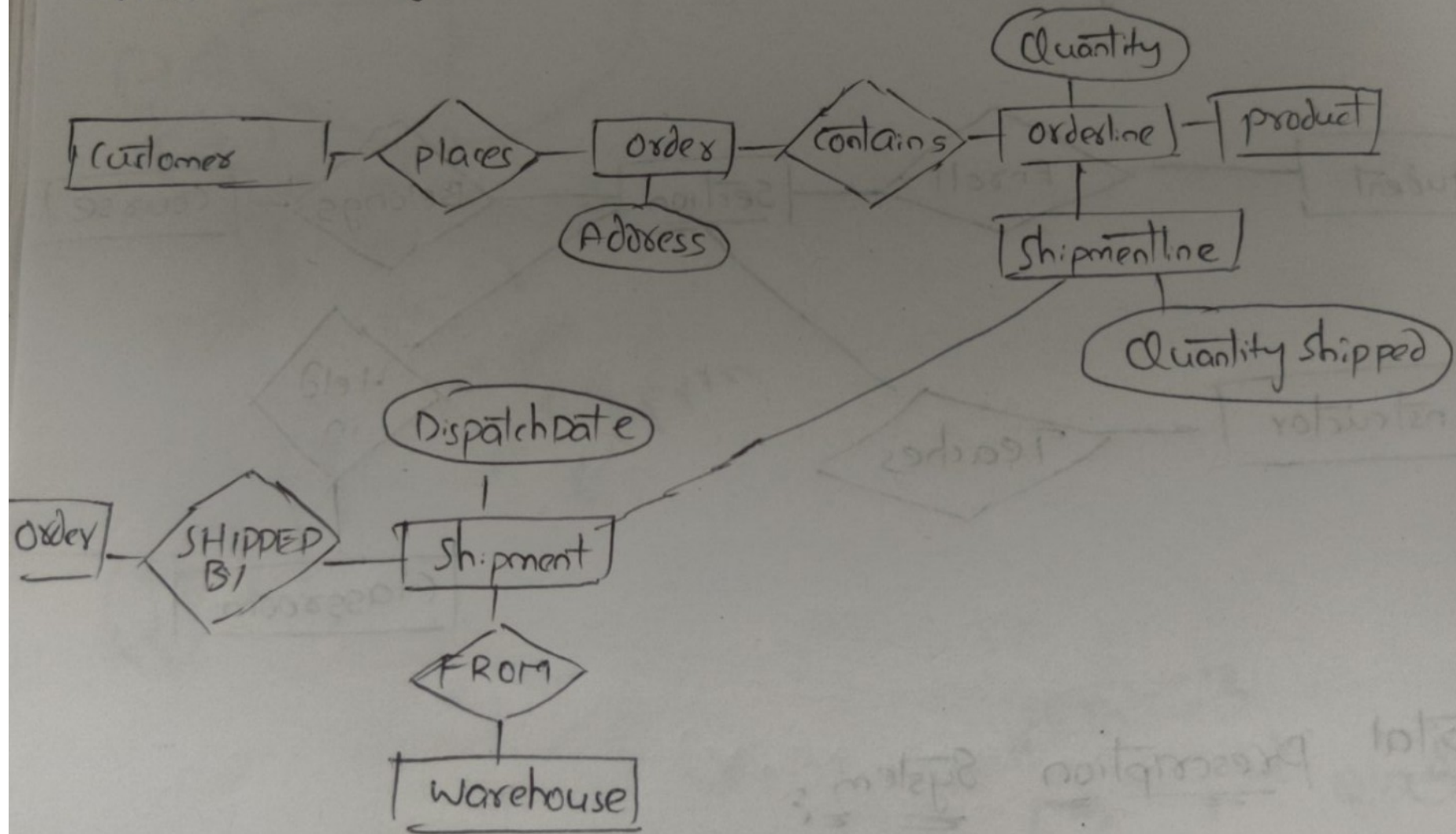
Flaws:

The “OrderLine” entity is missing and is required as quantity is specific to that orderline and not the product. Also other entity “ShipmentLine” is required for splitting the shipment.

1. Address should be an entity for Order and not the Customer as the customer might use different address for two different orders.
2. OrderLine entity is required as the quantity is entitled to the OrderLine. So, Order----CONTAINS-----OrderLine-----Product
3. ShipmentLine entity is required for keeping the track of patially shipped order line.
4. Keep Shipment with its own Dispatch date so that multiple shipments can be allowed per order

3) E-commerce Order Fulfilment

Correct ER-Diagram:



4. Project Staffing and Roles

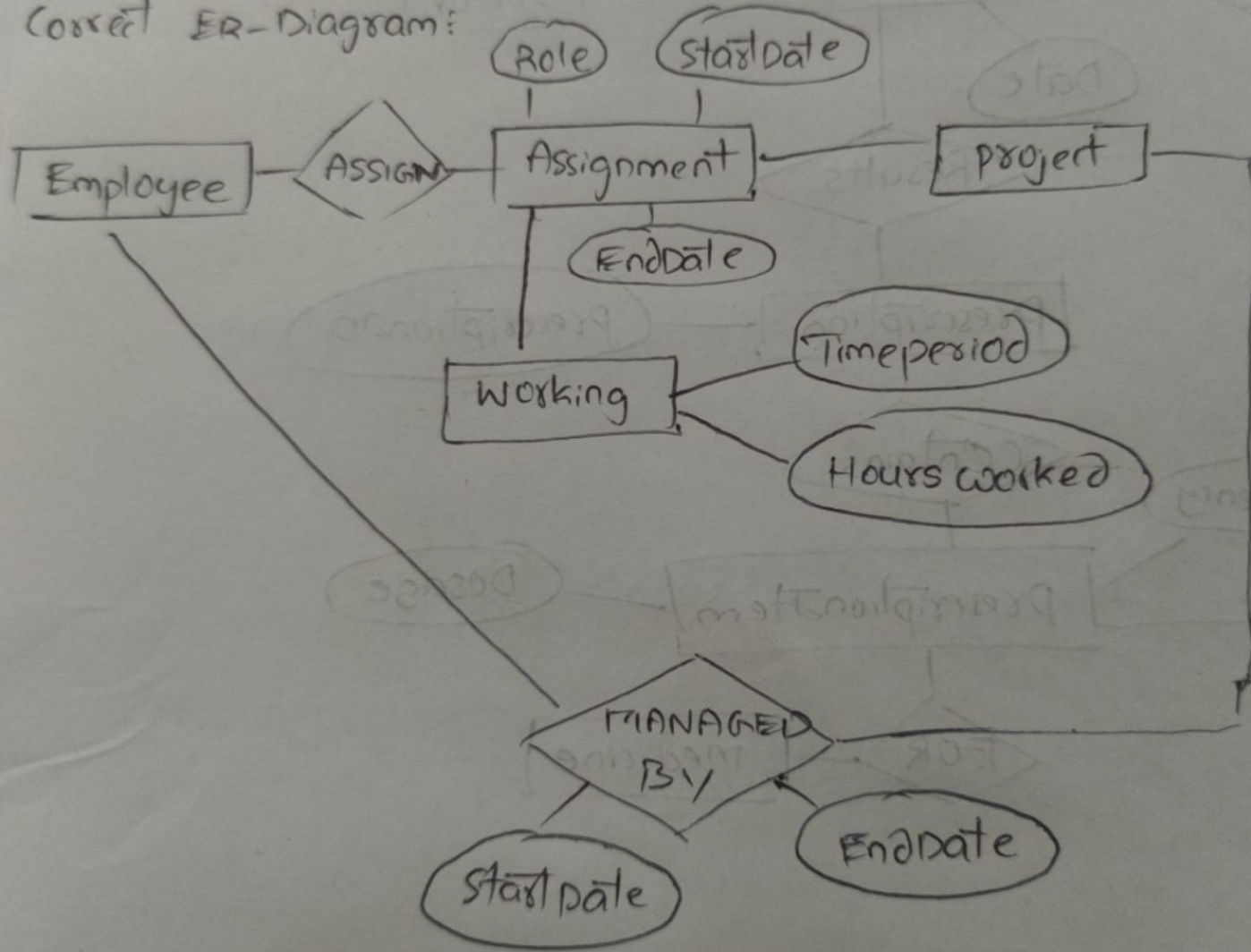
Flaws:

The relationship between Employee and Project should also be made an entity. By doing this, we would be able to manage the Employee-Project history.

1. There should be an Assignment entity which stores the role of that particular employee in a project.
2. Again, the Works_on relationship cannot represent joining, leaving and rejoining. So, Assignment should be entitled with Start and End date.
3. Since hours are recorded per employee-project assignment. So, it can't be stored in the Project entity neither in Assignment because It requires hours per time period, Hence, we can make a new entity WorkLogs which stores hours worked per time period per assignment.
4. We should keep Start and End date as attributes for the MANAGED_BY relationship as managers might change over a period of time.

4) Project staffing and Roles:

Correct ER-Diagram:



5. Airline Booking and Seat Assignment

Flaws:

The current ER diagram does not properly represent bookings, flight occurrences, aircraft changes, and seat assignments.

1. There should be a PNR entity because one PNR can contain multiple passengers and multiple flight segments.
2. There should be a FlightOccurrence entity because same flight number can operate on different dates.
3. Aircraft entity should be connected to FlightOccurrence because the aircraft assigned to a flight may change for different occurrences.
4. SeatAssignment entity should be there to store different seats for the same passenger on different flight segments.

Question 5 – Airline Booking and Seat Assignment

